Lab 2

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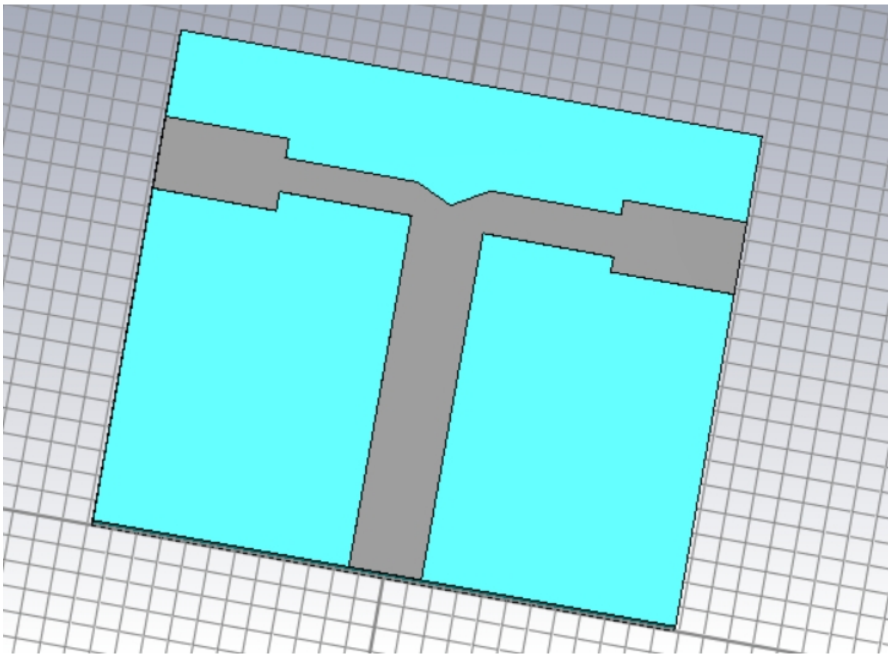
A)

We calculate the parameters:

Parameter	Value
l_2	7.2546 mm
l_3	7.1973 mm
w_2	1.3725 mm
w_3	1.7283 mm
w_0	2.9384 mm
k	7.1

and implement the T-junction in CST:

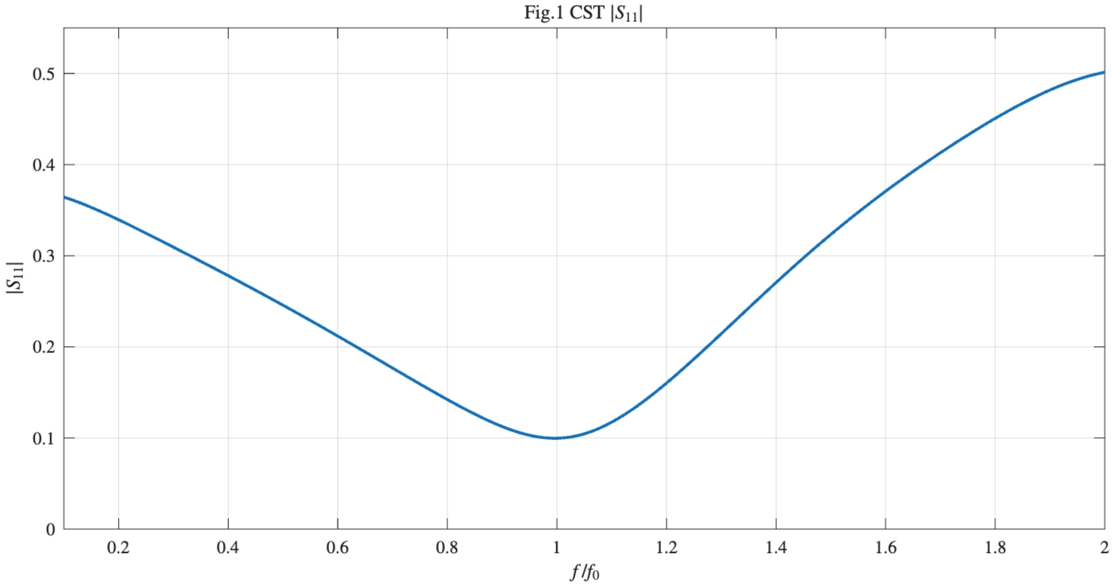


Then we tune the parameters to adjust the central frequency.

Parameter	Original (mm)	Tuned (mm)
l_2	7.2546	6.8193
l_3	7.1973	6.7655

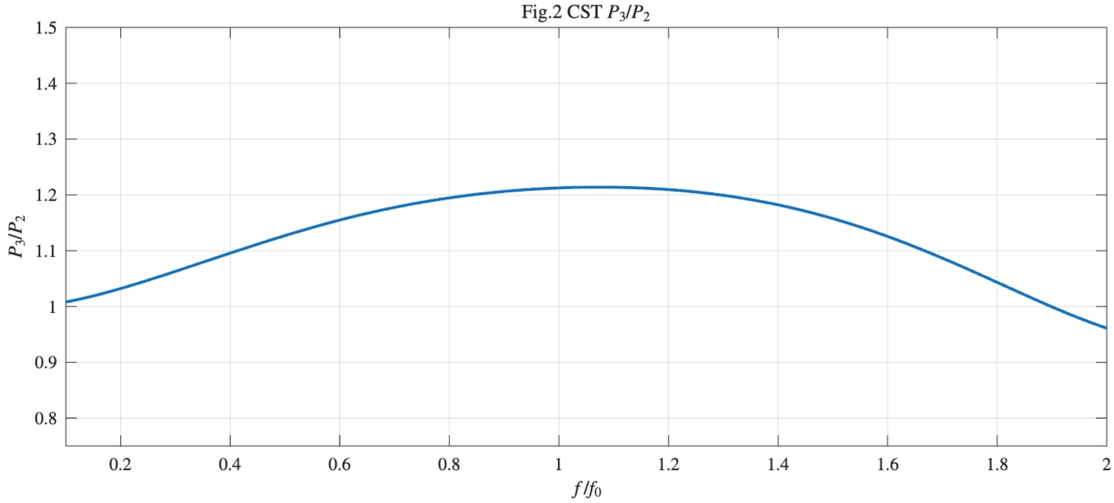
We run the solver and get the results as follow:

Fig.1 CST $|S_{11}|$



The junction is best matched near the design frequency.

Fig.2 CST P_3/P_2

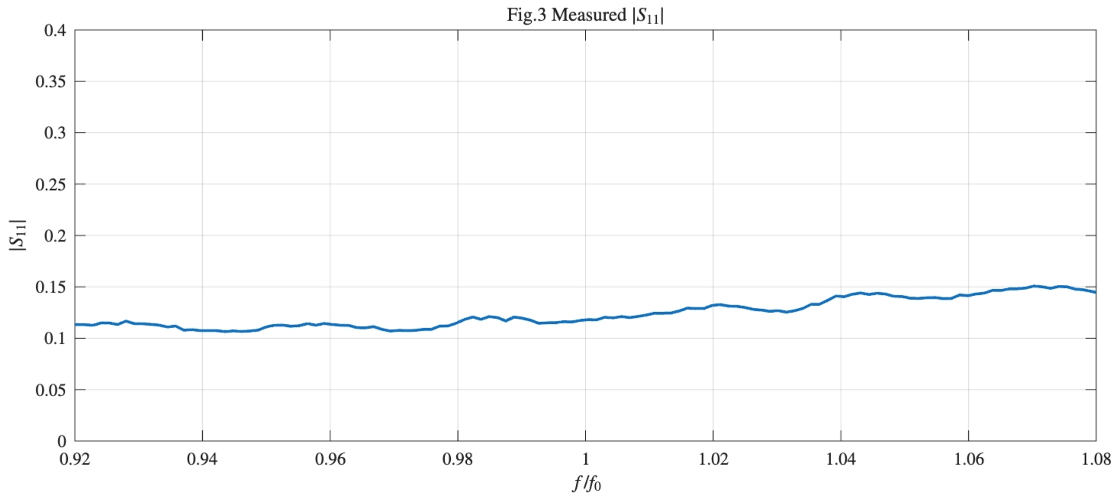


The power ratio P_3/P_2 is closest to the design value near $f/f_0=1$. Away from the center frequency, the ratio changes gradually, so the power division is frequency-dependent and is only most accurate around the design frequency.

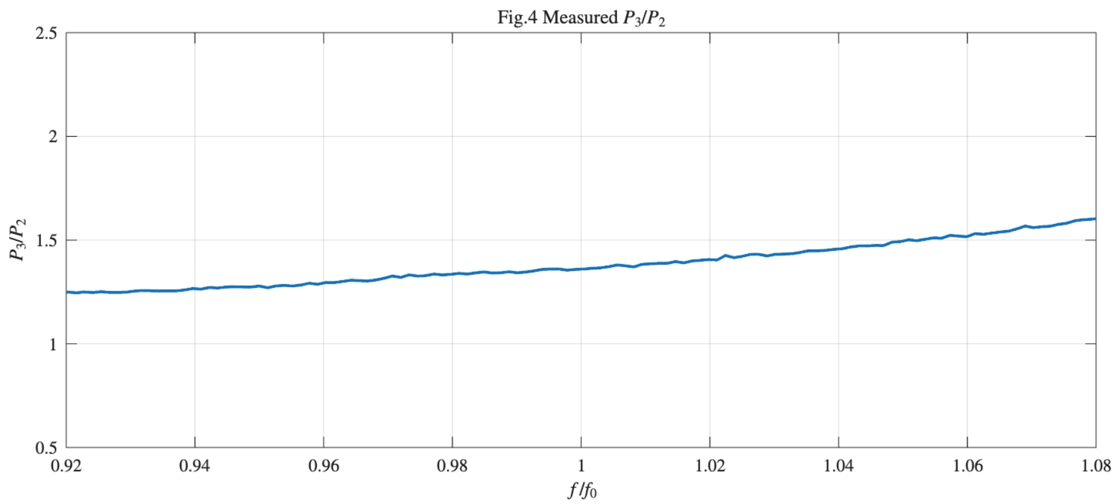
B)

First, a calibration procedure was performed on the VNA in the frequency range of interest. After calibration, the T-junction was measured. Since the device is a three-port structure and the VNA is a two-port instrument, the measurement was carried out in two steps. In the first step, one output branch was connected to the VNA together with port 1, while the other output branch was terminated with a 50Ω load, in order to measure S_{11} and one transmission coefficient. In the second step, the other output branch was connected, and the remaining branch was again terminated with a 50Ω load, so that the other transmission coefficient could be measured. In this way, the measured data were used to evaluate S_{11} and the power ratio P_3/P_2.

We use Matlab to read the s2p files and plot the figures as below:



The measured |S11| is around 0.11 to 0.15 in the whole range. This means the device is not perfectly matched, but the input reflection is still acceptable. The curve changes slightly, so we can say that the matching is relatively stable near the design frequency.

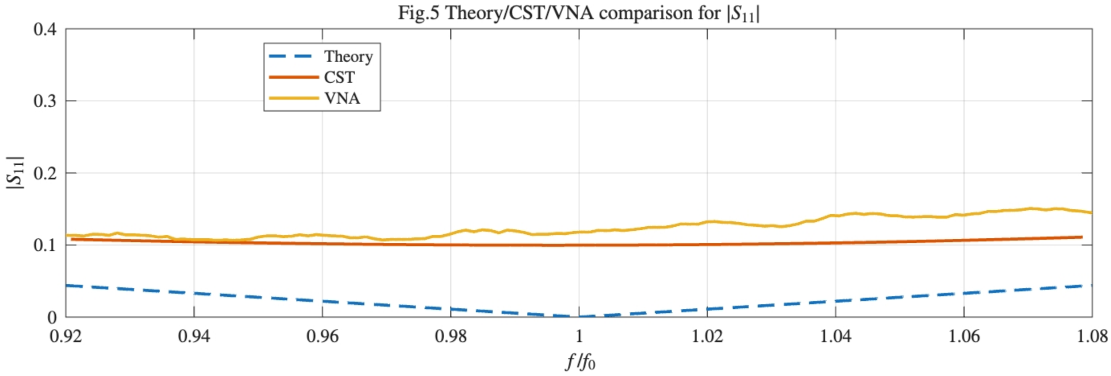


The measured power ratio P3/P2 is always higher than 1.25 and increases with frequency. This means port 3 receives more power than port 2 in the measurement, and the power division is not exactly the ideal design value. The difference may come from fabrication error, connector effect, and measurement condition.

C)

We use Matlab to combine the results of homework 2, VNA experiment and CST simulation. The figure is shown as below:

Fig.5 Theory/CST/VNA comparison for $|S_{11}|$



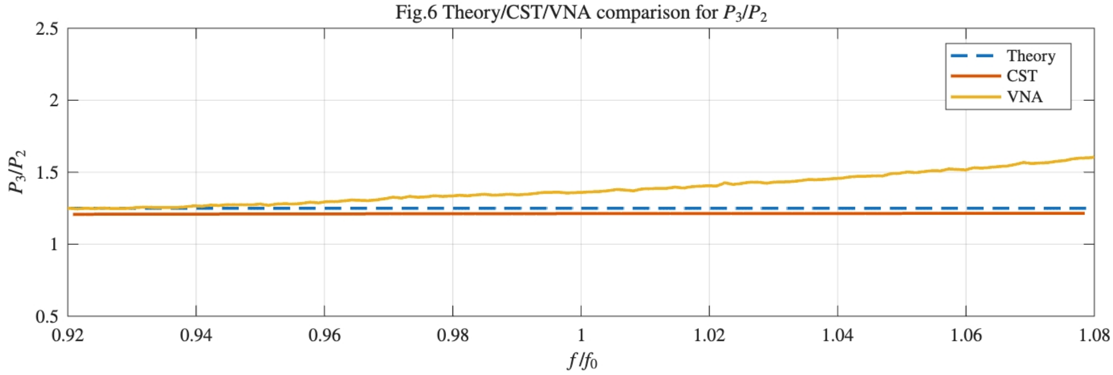
For Fig. 5, the theoretical curve has the lowest $|S_{11}|$ and reaches its minimum at $f/f_0=1$, because it is based on the ideal transmission-line model.

The CST result is higher than the theoretical one, since the electromagnetic simulation includes the practical microstrip realization and the discontinuity of the T-junction.

The VNA curve is the highest and shows small fluctuations. This means the measured device has more mismatch than both the theory and the CST model. The difference may be caused by fabrication tolerance, connector and soldering effects, material uncertainty, measurement errors and so on.

D)

Fig.6 Theory/CST/VNA comparison for P_3/P_2



For Fig. 6, the theoretical curve is constant at $P_3/P_2=1.25$, which is the ideal design target.

The CST result is slightly lower than the theoretical value, showing that the practical microstrip implementation does not exactly follow the ideal model.

The VNA result is higher than both theory and CST, and it increases with frequency. This means that in the measurement port 3 receives more power than expected, and the power division changes with frequency. The difference may come from fabrication error, junction discontinuity, connector effects, and measurement uncertainty.

Conclusion

This lab studied the design, simulation and measurement of a microstrip T-junction. First, the structure was designed and analyzed theoretically. Then the corresponding geometry was implemented in CST and the frequency behaviour of $|S_{11}|$ and P_3/P_2 was evaluated. After that, the device was characterized experimentally with the VNA.

From the comparison, the theoretical results represent the ideal target, while the CST results include the effect of the practical microstrip implementation. The measured results show larger deviations, mainly due to fabrication tolerances, connector and soldering effects, and measurement uncertainties. In particular, the measured $|S_{11}|$ is higher than the theoretical value, and the measured P_3/P_2 is also larger than the ideal design value.

Overall, the lab shows that the T-junction works as expected around the design frequency, while differences between theory, simulation and measurement appear because of practical non-ideal effects.